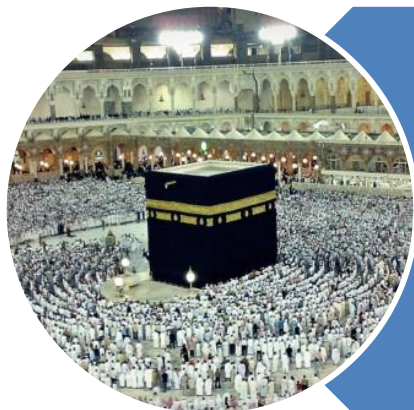


رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

Graph

Traversal

C

P

C

S

2

0

4

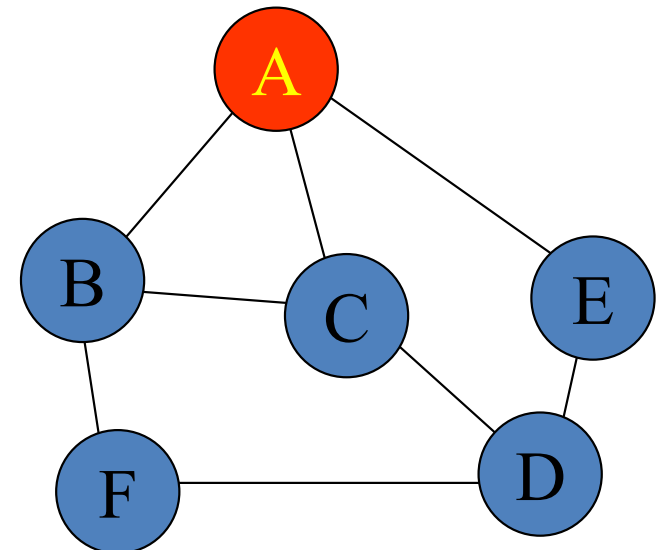
3

Graph - **Traversal**

- Depth First Search (**DFS**) Traversal
- Breadth First Search (**BFS**) Traversal

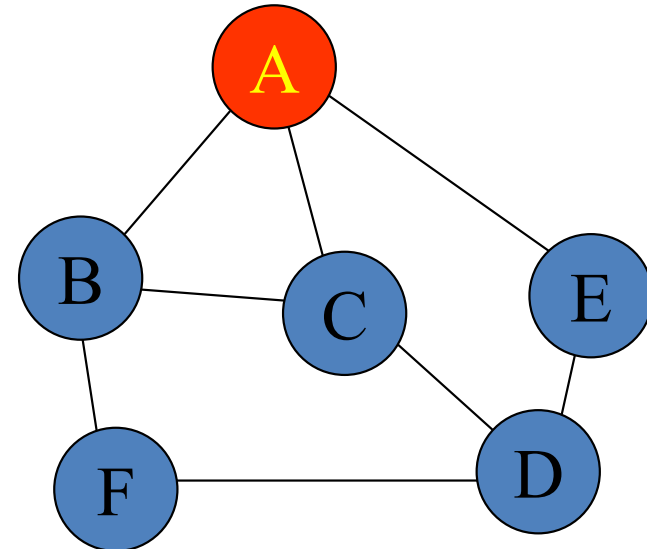
Traversal - DFS

- Probing List is implemented as **stack** (LIFO)
- Example
 - A's neighbor: B, C, E
 - B's neighbor: A, C, F
 - C's neighbor: A, B, D
 - D's neighbor: E, C, F
 - E's neighbor: A, D
 - F's neighbor: B, D
 - **Start from vertex A**



Traversal - DFS

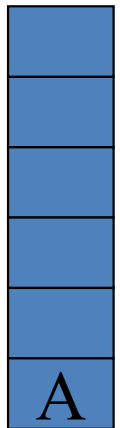
- A's neighbor: B C E
- B's neighbor: A C F
- C's neighbor: A B D
- D's neighbor: E C F
- E's neighbor: A D
- F's neighbor: B D



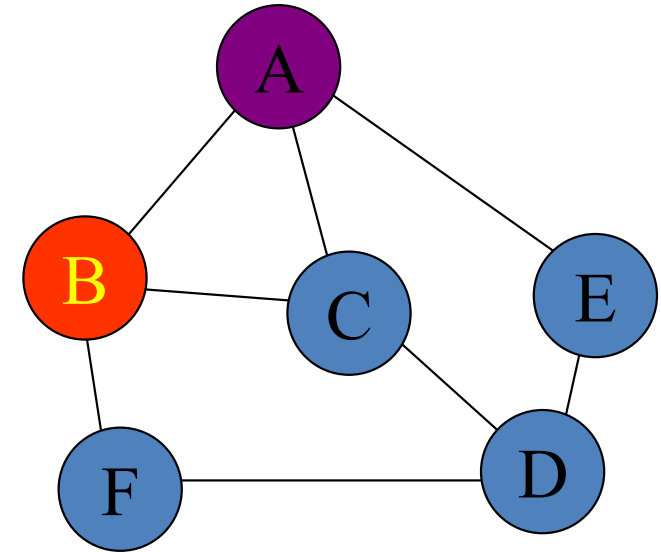
• Initial State

- Visited Vertices { }
- Probing Vertices { **A** }
- Unvisited Vertices { A, B, C, D, E, F }

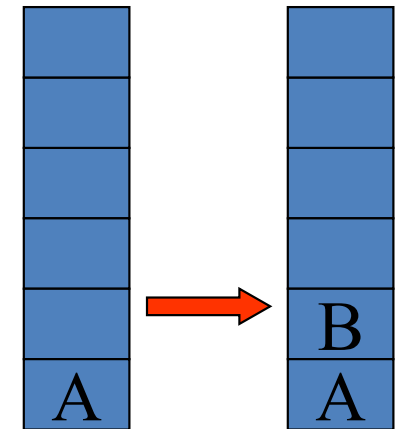
stack



Traversal - DFS

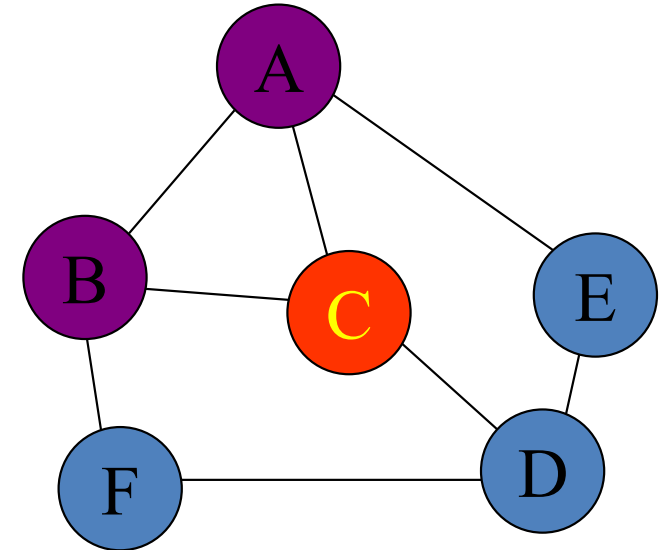


- Pick a vertex from stack, it is A, mark it as visited
- Find A's first unvisited neighbor, push it into stack
 - Visited Vertices { A }
 - Probing vertices { A, B }
 - Unvisited Vertices { B, C, D, E, F }

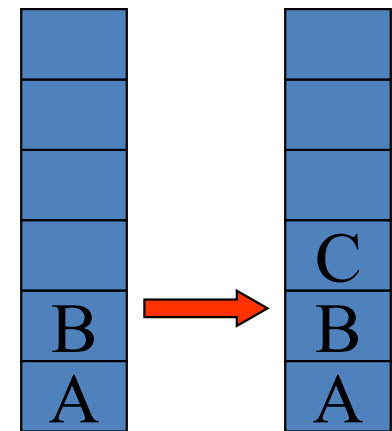


stack

Traversal - DFS

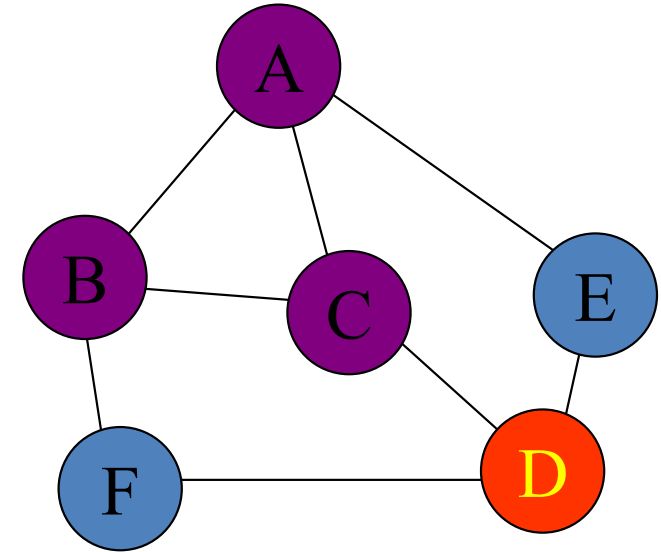


- Pick a vertex from stack, it is B, mark it as visited
- Find B's first unvisited neighbor, push it in stack
 - Visited Vertices { A, B }
 - Probing Vertices { A, B, C }
 - Unvisited Vertices { C, D, E, F }

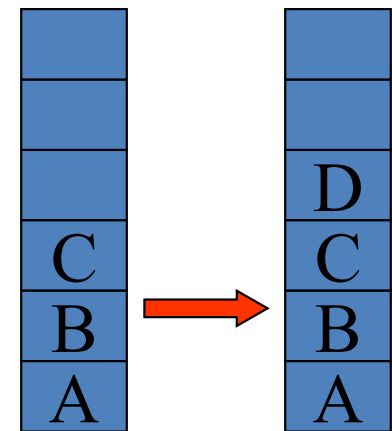


stack

Traversal - DFS

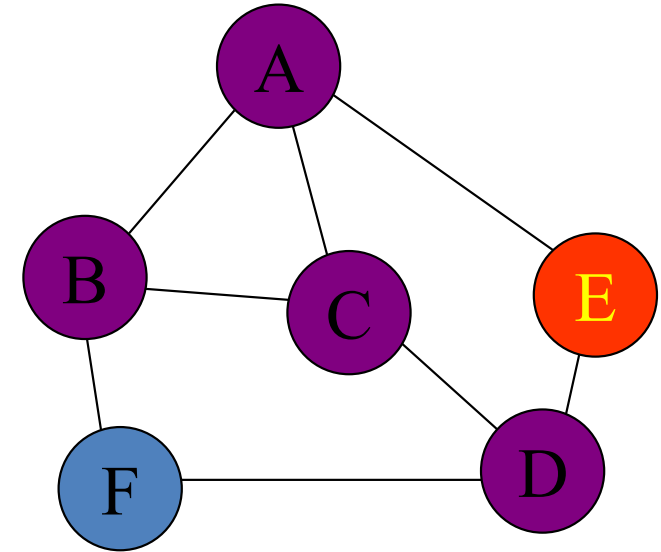


- Pick a vertex from stack, it is C, mark it as visited
- Find C's first unvisited neighbor, push it in stack
 - Visited Vertices { A, B, C }
 - Probing Vertices { A, B, C, D }
 - Unvisited Vertices { D, E, F }

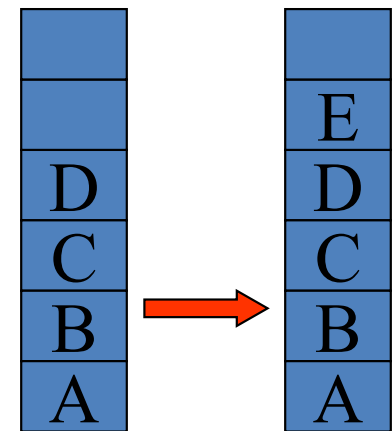


stack

Traversal - DFS

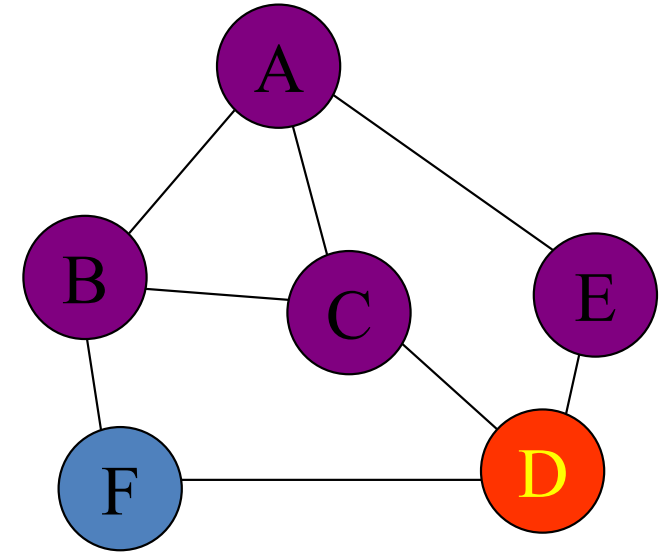


- Pick a vertex from stack, it is D, mark it as visited
- Find D's first unvisited neighbor, push it in stack
 - Visited Vertices { A, B, C, D }
 - Probing Vertices { A, B, C, D, E }
 - Unvisited Vertices { E, F }

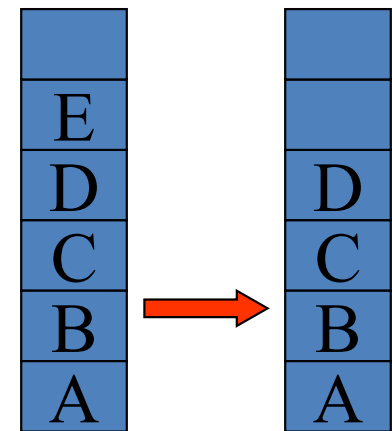


stack

Traversal - DFS

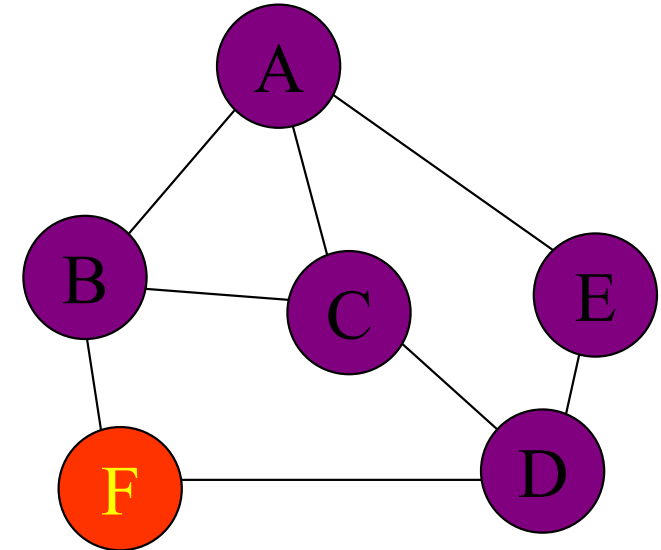


- Pick a vertex from stack, it is E, mark it as visited
- Find E's first unvisited neighbor, no vertex found, Pop E
 - Visited Vertices { A, B, C, D, E }
 - Probing Vertices { A, B, C, D }
 - Unvisited Vertices { F }

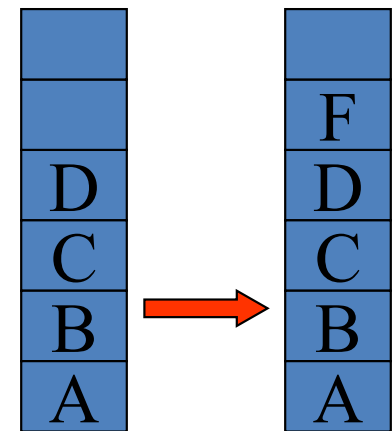


stack

Traversal - DFS

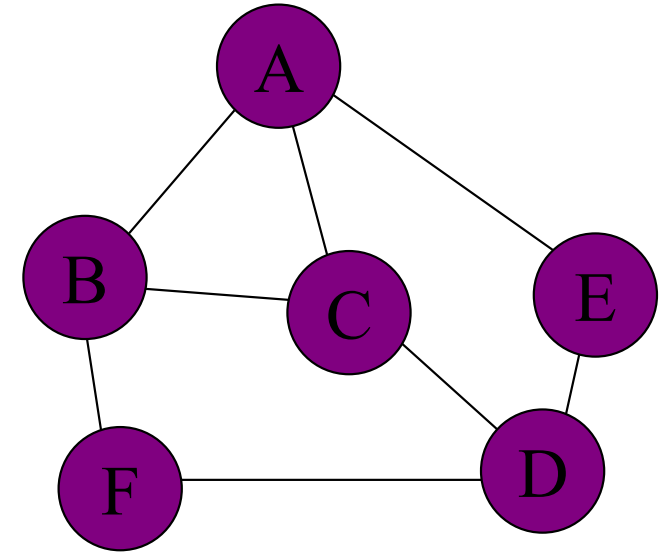


- Pick a vertex from stack, it is D, mark it as visited
- Find D's first unvisited neighbor, push it in stack
 - Visited Vertices { A, B, C, D, E }
 - Probing Vertices { A, B, C, D, F }
 - Unvisited Vertices { F }

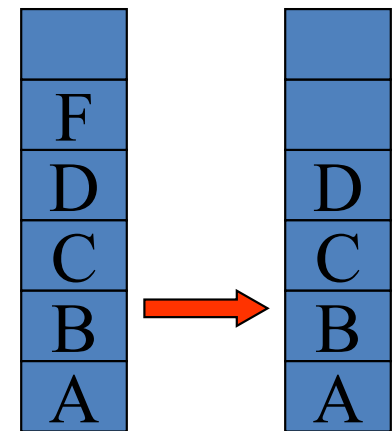


stack

Traversal - DFS

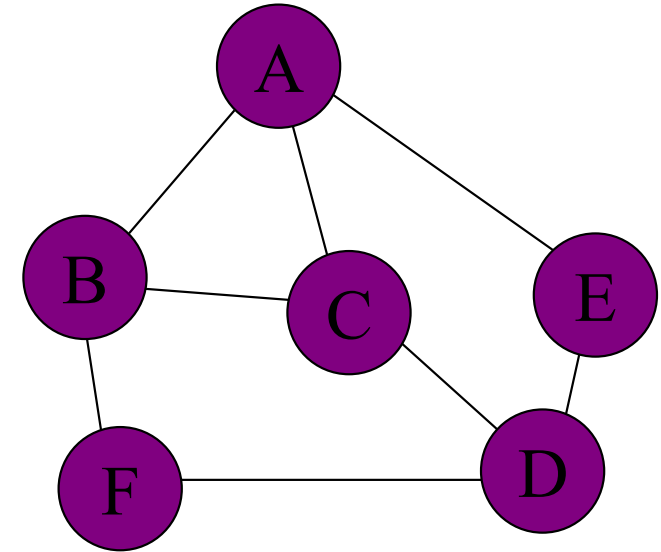


- Pick a vertex from stack, it is F, mark it as visited
- Find F's first unvisited neighbor, no vertex found, Pop F
 - Visited Vertices { A, B, C, D, E, F }
 - Probing Vertices { A, B, C, D }
 - Unvisited Vertices { }

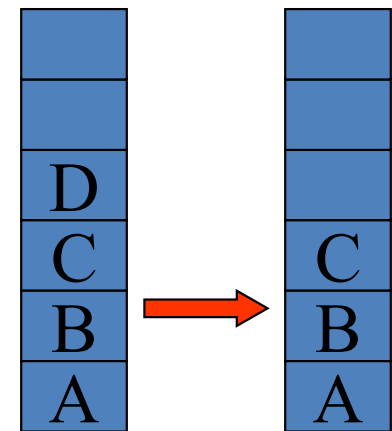


stack

Traversal - DFS

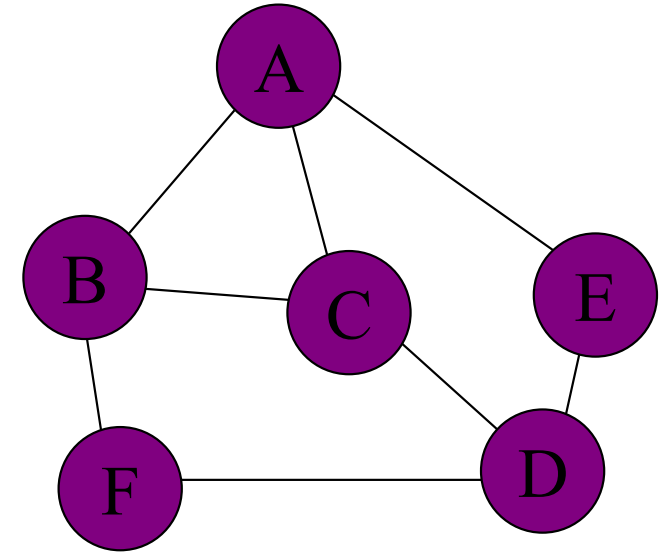


- Pick a vertex from stack, it is D, mark it as visited
- Find D's first unvisited neighbor, no vertex found, Pop D
 - Visited Vertices { A, B, C, D, E, F }
 - Probing Vertices { A, B, C }
 - Unvisited Vertices { }

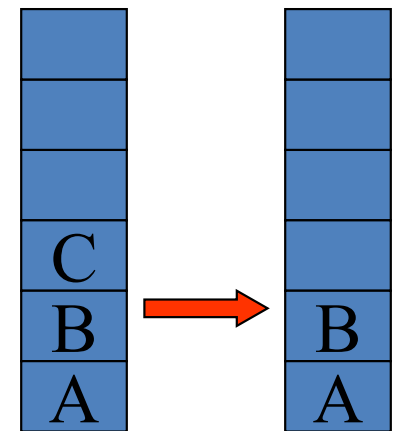


stack

Traversal - DFS

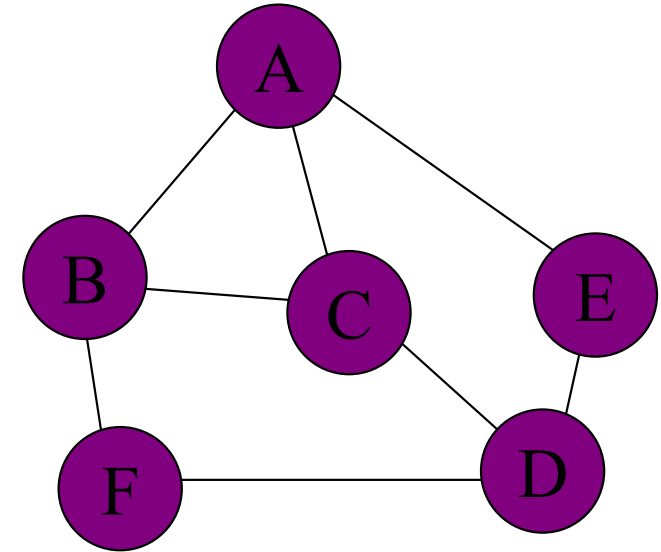


- Pick a vertex from stack, it is C, mark it as visited
- Find C's first unvisited neighbor, no vertex found, Pop C
 - Visited Vertices { A, B, C, D, E, F }
 - Probing Vertices { A, B }
 - Unvisited Vertices { }

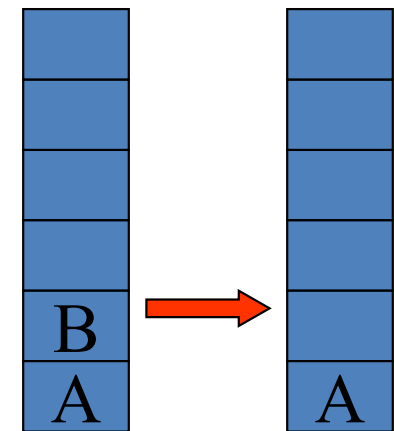


stack

Traversal - DFS

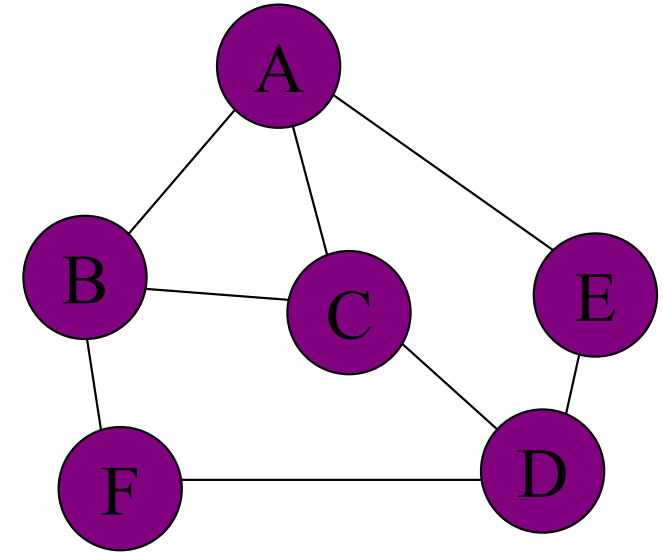


- Pick a vertex from stack, it is B, mark it as visited
- Find B's first unvisited neighbor, no vertex found, Pop B
 - Visited Vertices { A, B, C, D, E, F }
 - Probing Vertices { A }
 - Unvisited Vertices { }

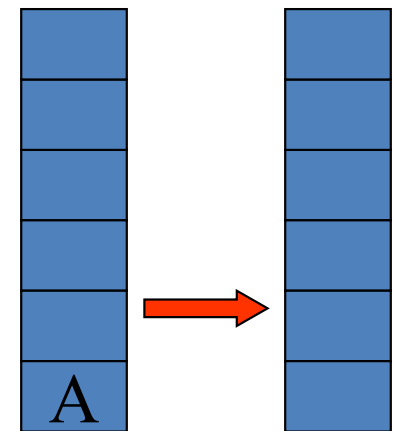


stack

Traversal - DFS

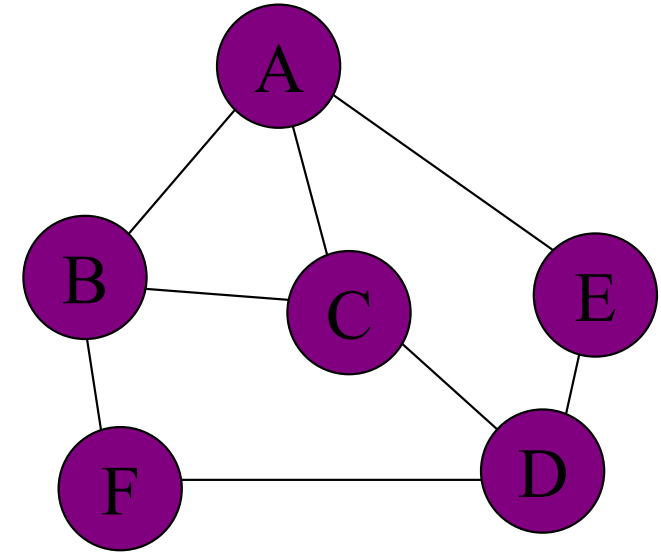


- Pick a vertex from stack, it is A, mark it as visited
- Find A's first unvisited neighbor, no vertex found, Pop A
 - Visited Vertices { A, B, C, D, E, F }
 - Probing Vertices { }
 - Unvisited Vertices { }



stack

Traversal - DFS

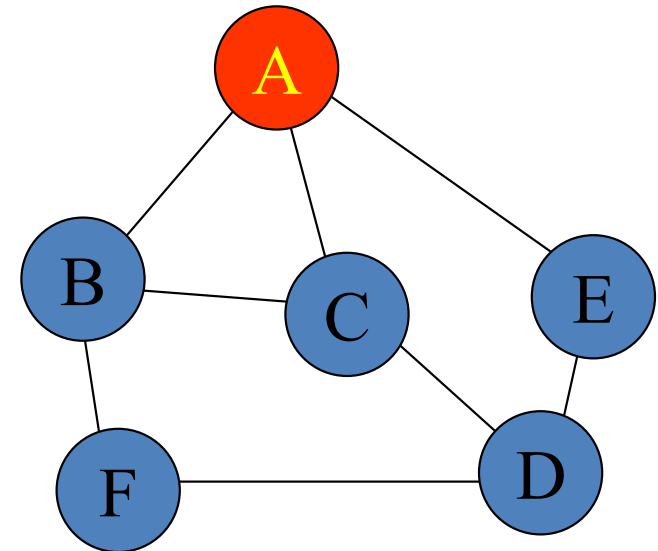


stack

- Now probing list is empty
- End of Depth First Traversal
 - Visited Vertices { A, B, C, D, E, F }
 - Probing Vertices { }
 - Unvisited Vertices { }

Traversal – Breadth First Search

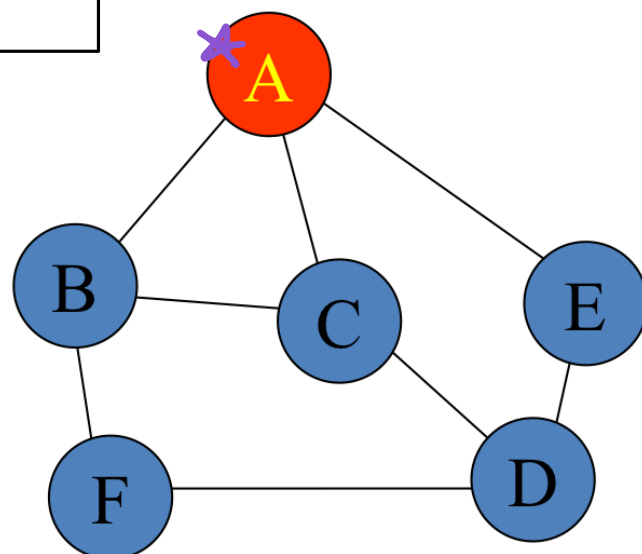
- Probing List is implemented as queue (FIFO)
- Example
 - A's neighbor: B C E
 - B's neighbor: A C F
 - C's neighbor: A B D
 - D's neighbor: E C F
 - E's neighbor: A D
 - F's neighbor: B D
 - **Start from vertex A**



	0	1	2	3	4	5	6
Probing Vertices							

unvisited

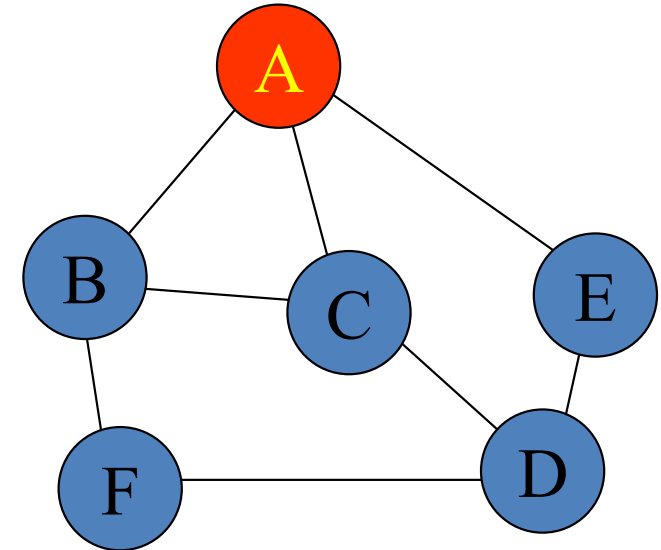
- ✓ A
- ✓ B
- ✓ C
- ✓ D
- ✓ E
- ✓ F



output

A B C E F D

Traversal – **BFS**



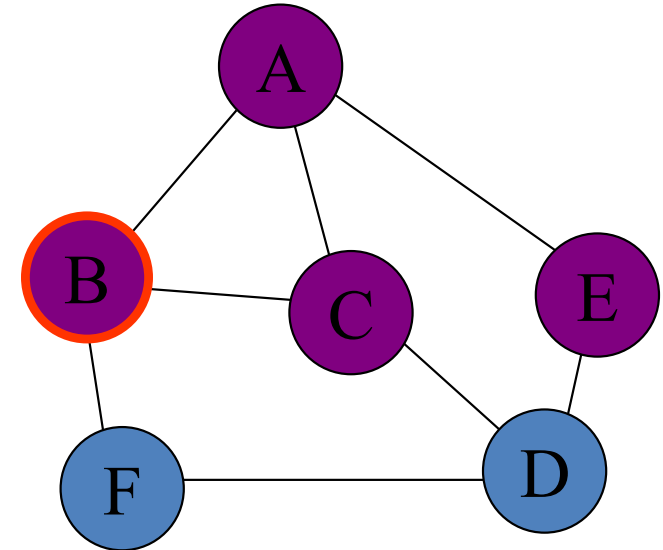
- Initial State

- Visited Vertices { }
- Probing Vertices { A }
- Unvisited Vertices { A, B, C, D, E, F }

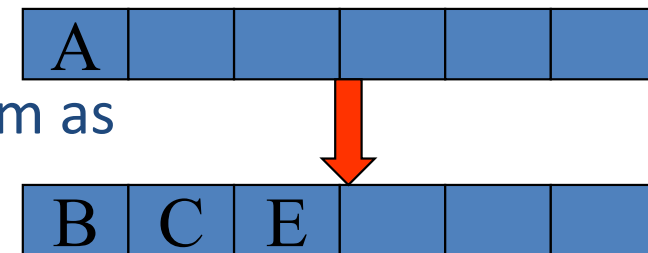


queue

Traversal – **BFS**

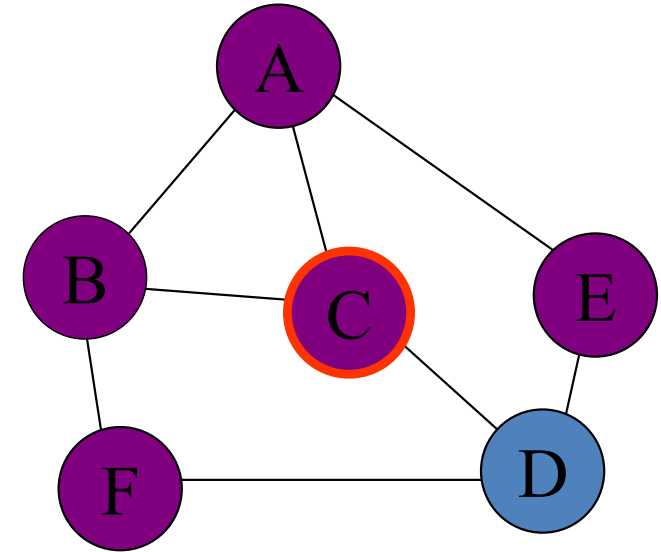


- Delete first vertex from queue, it is A, mark it as visited
- Find A's all unvisited neighbors, mark them as visited, put them into queue
 - Visited Vertices { A, B, C, E }
 - Probing Vertices { B, C, E }
 - Unvisited Vertices { D, F }

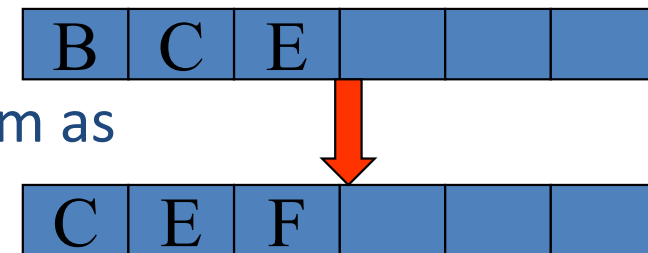


queue

Traversal – **BFS**

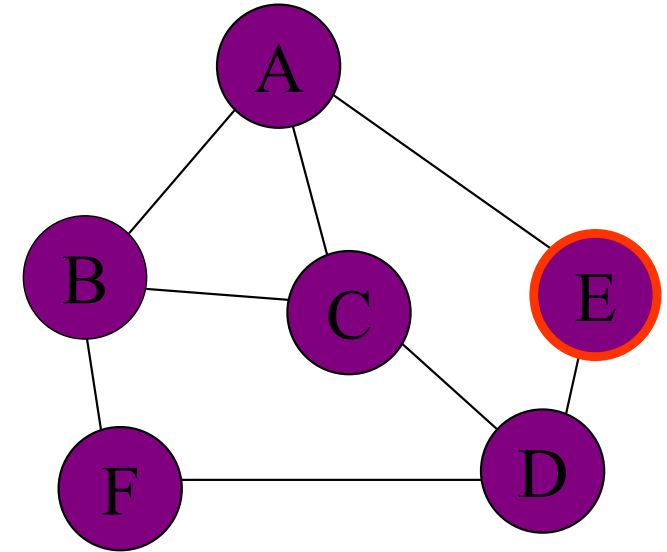


- Delete first vertex from queue, it is B, mark it as visited
- Find B's all unvisited neighbors, mark them as visited, put them into queue
 - Visited Vertices { A, B, C, E, F }
 - Probing Vertices { C, E, F }
 - Unvisited Vertices { D }



queue

Traversal – **BFS**

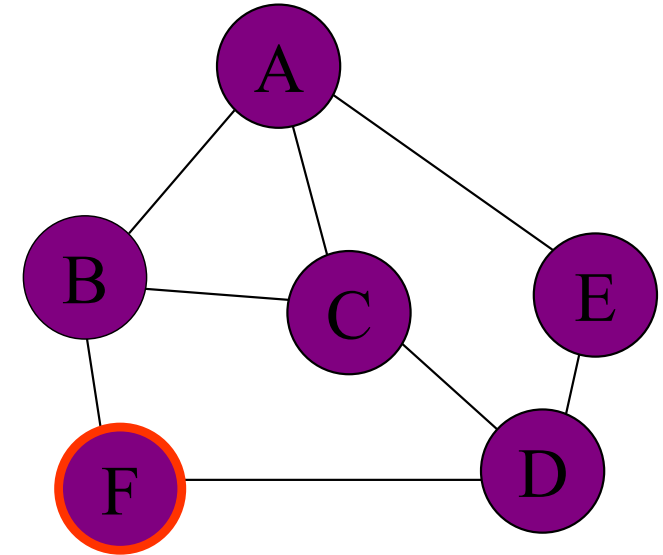


- Delete first vertex from queue, it is C, mark it as visited
- Find C's all unvisited neighbors, mark them as visited, put them into queue
 - Visited Vertices { A, B, C, E, F, D }
 - Probing Vertices { E, F, D }
 - Unvisited Vertices { }



queue

Traversal – BFS

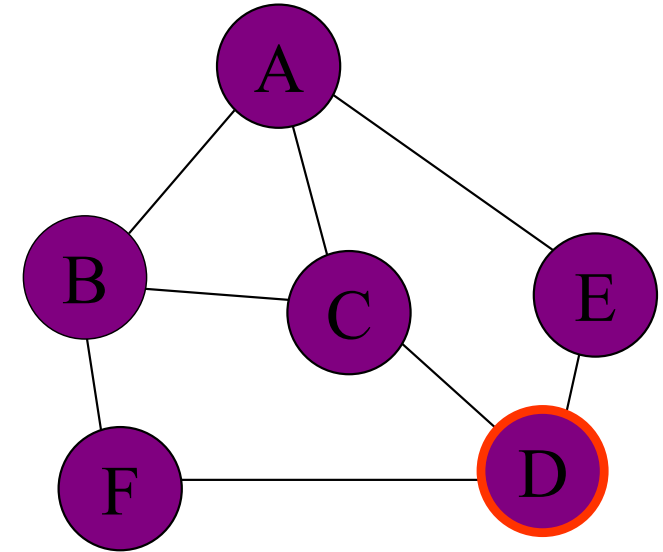


- Delete first vertex from queue, it is E, mark it as visited
- Find E's all unvisited neighbors, no vertex found
 - Visited Vertices { A, B, C, E, F, D }
 - Probing Vertices { F, D }
 - Unvisited Vertices { }



queue

Traversal – **BFS**

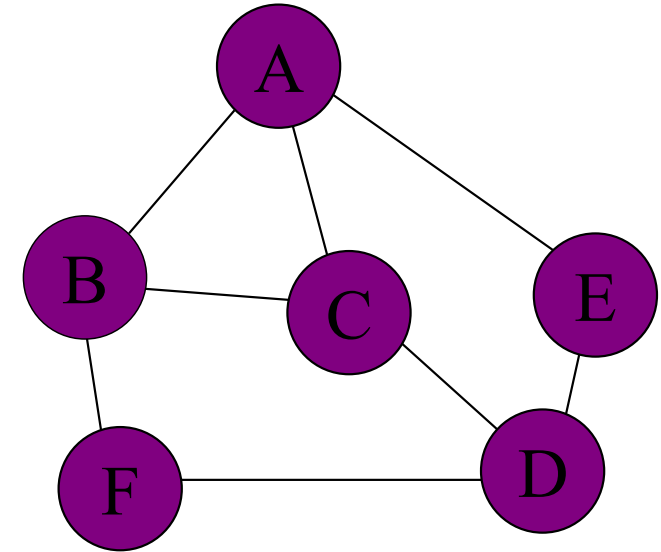


- Delete first vertex from queue, it is F, mark it as visited
- Find F's all unvisited neighbors, no vertex found
 - Visited Vertices { A, B, C, E, F, D }
 - Probing Vertices { D }
 - Unvisited Vertices { }



queue

Traversal – BFS

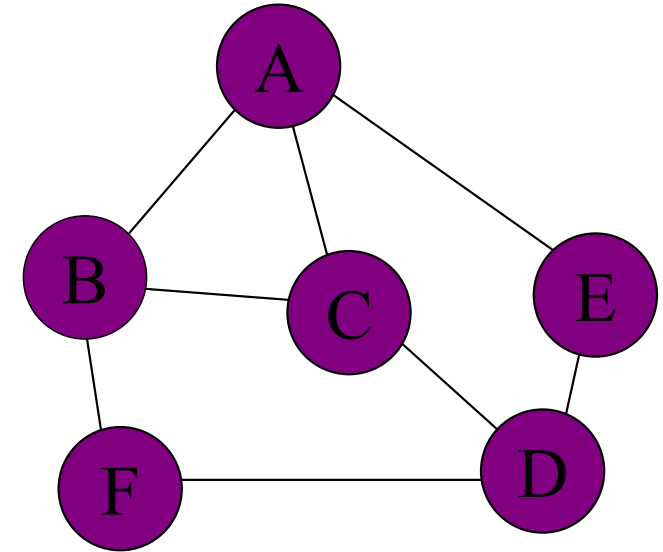


- Delete first vertex from queue, it is D, mark it as visited
- Find D's all unvisited neighbors, no vertex found
 - Visited Vertices { A, B, C, E, F, D }
 - Probing Vertices { }
 - Unvisited Vertices { }



queue

Traversal – BFS



- Now the queue is empty
- End of Breadth First Traversal
 - Visited Vertices { A, B, C, E, F, D }
 - Probing Vertices { }
 - Unvisited Vertices { }

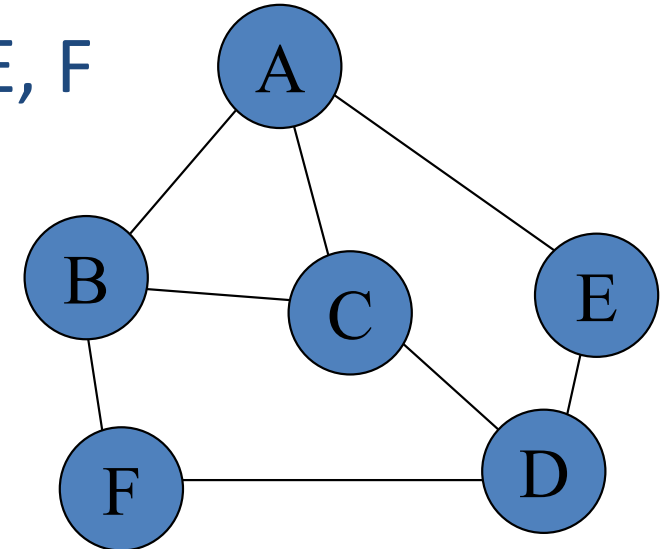


queue

Difference – DFS & BFS

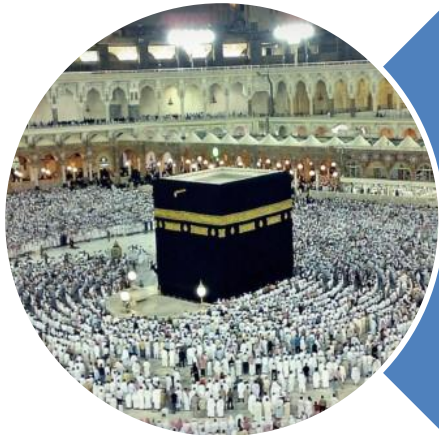
- Depth First Search (DFS)**

- Order of visited: A, B, C, D, E, F



- Breadth First Search (BFS)**

- Order of visited: A, B, C, E, F, D



Allah (Alone) is Sufficient for us,
and He is the Best Disposer of
affairs (for us). **(Quran 3 : 173)**